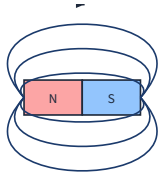
# CHAPTER 17 MAGNETIC FIELDS

**Key idea:** moving charges and currents create magnetic fields and feel forces in them.

## 1 MAGNETIC FIELD

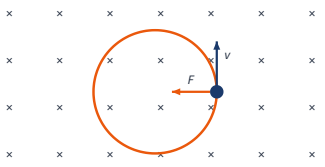


field lines from N to S outside the magnet

$$B = \frac{F}{IL\sin\theta}$$

- 1 T = 1 N A$^{-1}$ m$^{-1}$
- Lines: closed loops, N → S outside a magnet
- Right-hand grip rule for wires and coils

## 2 FORCE ON A MOVING CHARGE



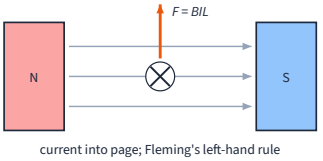
B into page: F = qvB towards centre, r = mv/qB

$$F = qvB\sin\theta$$

$$r = \frac{mv}{qB} \qquad T = \frac{2\pi m}{qB}$$

Force $\perp v$: does no work, speed constant. $T$ independent of $v$.

## 3 FORCE ON A CONDUCTOR



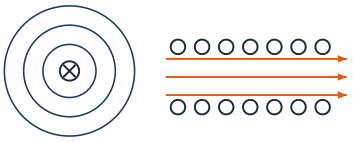
current into page; Fleming's left-hand rule

$$F = BIL\sin\theta$$

$$\tau = NBIA\sin\phi$$

**Fleming's left-hand rule:** first finger = field, second finger = current, thumb = force.

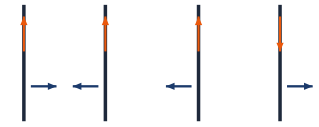
## 4 FIELDS DUE TO CURRENTS


wire: B = μ₀I/2πr solenoid: B = μ₀nI (uniform inside)

| Source | $B$ |
|---|---|
| Long straight wire | $\frac{\mu_0 I}{2\pi r}$ |
| Centre of coil | $\frac{\mu_0 NI}{2r}$ |
| Long solenoid | $\mu_0 nI$ |

$\mu_0 = 4\pi \times 10^{-7}$ H m$^{-1}$; $n$ = turns per metre.

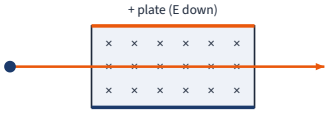
## 5 PARALLEL WIRES


same direction: attract opposite: repel

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

- Same direction: **attract**; opposite: repel
- Equal and opposite forces (3rd law)
- Defines the ampere: $2 \times 10^{-7}$ N m$^{-1}$ at 1 m

## 6 E/M & VELOCITY SELECTOR



B into page
undeflected when eE = evB → v = E/B

$$v = \frac{E}{B} \qquad \frac{e}{m} = \frac{2V}{B^2 r^2}$$

Crossed fields: undeflected when $qE = qvB$. $e/m = 1.76 \times 10^{11}$ C kg$^{-1}$.

## 7 HALL EFFECT

$$V_H = \frac{BI}{nte}$$

- Carriers pushed to one side until $eE_H = evB$
- Sign of $V_H$ shows the sign of the carriers
- Semiconductor: small $n$, large $V_H$ (Hall probe)

## 8 COMMON SECTION A TRAPS

- For electrons, reverse the left-hand rule.
- Using $n$ as total turns in $B = \mu_0 nI$.
- Saying the magnetic force changes the speed.
- $t$ in $V_H$ is the thickness **along B**.
- Angle $\theta$ is between the wire (or $v$) and $B$.

## EXAM TIPS

- ✓Draw $B$, $I$ (or $v$) and $F$ as a 3-D sketch.
- ✓Write "magnetic force provides centripetal force".
- ✓Add fields from two wires as vectors.
- ✓Check units: T, A, m.